

Aircraft Flight Crash Simulator

Unit Test Report

Generated: 2026-05-10 22:56:51

ALL TESTS PASSED ✓

Total: 226 | Passed: 226 | Failed: 0

Test Details

General**[226/228]**

- ✓ PHASE_ON_GROUND name
- ✓ PHASE_TAKEOFF name
- ✓ PHASE_CLIMB name
- ✓ PHASE_CRUISE name
- ✓ PHASE_DESCENT name
- ✓ PHASE_LANDING name
- ✓ PHASE_CRASH name
- ✓ PHASE_POST_CRASH name
- ✓ t=0 is ON_GROUND
- ✓ t=5 is ON_GROUND
- ✓ t=16 is TAKEOFF
- ✓ t=40 is CLIMB
- ✓ t=150 is CRUISE
- ✓ t=290 is DESCENT
- ✓ t=360 is LANDING
- ✓ crashed=1 ? CRASH
- ✓ crashed=1 at t=0 ? CRASH
- ✓ 200s: t=0 ON_GROUND
- ✓ 200s: t=10 TAKEOFF
- ✓ 200s: t=25 CLIMB
- ✓ 200s: t=80 CRUISE
- ✓ 200s: t=160 DESCENT
- ✓ 200s: t=185 LANDING
- ✓ ON_GROUND altitude = 0
- ✓ CRUISE altitude ~9000m
- ✓ TAKEOFF altitude > 0
- ✓ CLIMB altitude > 500m
- ✓ DESCENT altitude < 9000m
- ✓ LANDING altitude < 500m
- ✓ ON_GROUND speed = 0
- ✓ TAKEOFF speed = 75 m/s
- ✓ CRUISE speed = 232 m/s (451 kts)
- ✓ CLIMB speed > 75 m/s
- ✓ CLIMB speed <= 220 m/s
- ✓ ON_GROUND pitch=0
- ✓ TAKEOFF pitch=12

✓	CLIMB pitch=8
✓	CRUISE pitch=2
✓	DESCENT pitch=-3
✓	LANDING pitch=5
✓	TAKEOFF roll=0
✓	LANDING roll=0
✓	CRUISE roll=2
✓	CRASH_NONE name
✓	CRASH_STALL name
✓	CRASH_OVERSPEED name
✓	CRASH_NOSE_DIVE name
✓	CRASH_SPIRAL name
✓	CRASH_ENGINE_FAILURE name
✓	CRASH_TERRAIN name
✓	CrashInfo.type = STALL
✓	CrashInfo.trigger_time = 200
✓	CrashInfo.impact_time = trigger+30
✓	CrashInfo.description not empty
✓	CrashInfo.cause not empty
✓	CrashInfo NONE type
✓	STALL reduces altitude
✓	OVERSPEED reduces altitude
✓	NOSE_DIVE reduces altitude
✓	SPIRAL reduces altitude
✓	ENGINE_FAILURE reduces altitude
✓	TERRAIN reduces altitude
✓	STALL altitude >= 0
✓	OVERSPEED altitude >= 0
✓	TERRAIN faster than ENGINE_FAILURE
✓	SPIRAL faster than STALL
✓	At trigger moment, altitude unchanged
✓	STALL pitch decreases over time
✓	NOSE_DIVE pitch = -45 deg
✓	ENGINE_FAILURE pitch = -5 deg
✓	Random crash type in valid range
✓	Random crash type in valid range
✓	Random crash type in valid range

[illegible]

[illegible]

✓	Random crash type in valid range
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✓	Random crash type in valid range
✓	Random crash type in valid range
✓	Random crash type in valid range
✓	Some runs have no crash
✓	Some runs have a crash
■	In 100 trials: 53 normal, 47 crashes
✓	Empty after init
✓	Count = 5 after 5 inserts
✓	get_ordered_records returns 5
✓	get_ordered_records not NULL
✓	First record time = 0
✓	Last record time = 4
✓	Count capped at BLACKBOX_SIZE (200)
✓	Ordered count = 200
✓	Oldest remaining = record 10
✓	Newest remaining = record 209
✓	50 records returned

✓	Records are in chronological order
✓	20 records before reinit
✓	0 records after reinit
✓	get_ordered_records returns NULL when empty
✓	ON_GROUND altitude = 0
✓	CRUISE altitude near 9000m
✓	Altitude clamped to [0,10000]
✓	Altitude decreases during crash
✓	Altitude never negative during crash
✓	ON_GROUND speed = 0
✓	After 30 ticks CRUISE speed ~232 m/s (451 kts)
✓	Speed decreases when crashed
✓	Speed never negative when crashed
✓	0 m/s = 0 kts
✓	1 m/s = 1.944 kts
✓	220 m/s = ~427 kts
✓	100 m/s = ~194 kts
✓	Pitch within sensor limits [-10,15]
✓	NOSE_DIVE crash pitch = -45 deg
✓	TAKEOFF roll near 0
✓	Normal roll always in [-35, 35]
✓	Crash roll can exceed normal +-35 limits
✓	No error for valid sensor values
✓	No error message for valid values
✓	Error flag set for altitude > 10000
✓	Error message set for altitude > 10000
✓	Error flag set for speed > 300
✓	Error flag set for pitch > 45
✓	Error flag set for pitch < -45
✓	Error flag set for roll > 90
✓	Exact max boundary values ? no error
✓	Exact min boundary values ? no error
✓	Timestamp length = 8
✓	Timestamp has ':' at position 2
✓	Timestamp has ':' at position 5
✓	Hour digit 1 is numeric
✓	Hour digit 2 is numeric

✓	Minute tens digit is 0-5
✓	Minute units digit is numeric
✓	Second tens digit is 0-5
✓	Second units digit is numeric
■	Current timestamp: 22:56:55
✓	After reset: ON_GROUND altitude = 0
✓	After reset: ON_GROUND speed = 0

Raw Output

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+-----+
? AIRCRAFT FLIGHT CRASH SIMULATOR ? UNIT TESTS ?
+-----+

+-----+
? Flight Phase ? Names ?
+-----+
[PASS] PHASE_ON_GROUND name
[PASS] PHASE_TAKEOFF name
[PASS] PHASE_CLIMB name
[PASS] PHASE_CRUISE name
[PASS] PHASE_DESCENT name
[PASS] PHASE_LANDING name
[PASS] PHASE_CRASH name
[PASS] PHASE_POST_CRASH name

+-----+
? Flight Phase ? Transitions (380s flight)?
+-----+
[PHASES] Duration=380s | ON_GROUND:0-15s | TAKEOFF:15-34s | CLIMB:34-114s | CRUISE:114-266s |
DESCENT:266-334s | LANDING:334-380s
[PASS] t=0 is ON_GROUND
[PASS] t=5 is ON_GROUND
[PASS] t=16 is TAKEOFF
[PASS] t=40 is CLIMB
[PASS] t=150 is CRUISE
[PASS] t=290 is DESCENT
[PASS] t=360 is LANDING
[PASS] crashed=1 ? CRASH
[PASS] crashed=1 at t=0 ? CRASH

+-----+
? Flight Phase ? Transitions (200s flight)?
+-----+
[PHASES] Duration=200s | ON_GROUND:0-8s | TAKEOFF:8-18s | CLIMB:18-60s | CRUISE:60-140s | DESCENT:140-176s |
LANDING:176-200s
[PASS] 200s: t=0 ON_GROUND
[PASS] 200s: t=10 TAKEOFF
[PASS] 200s: t=25 CLIMB
[PASS] 200s: t=80 CRUISE
[PASS] 200s: t=160 DESCENT
[PASS] 200s: t=185 LANDING
[PHASES] Duration=380s | ON_GROUND:0-15s | TAKEOFF:15-34s | CLIMB:34-114s | CRUISE:114-266s |
DESCENT:266-334s | LANDING:334-380s

+-----+
? Flight Phase ? Target Altitude ?
+-----+
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[PHASES] Duration=380s | ON_GROUND:0-15s | TAKEOFF:15-34s | CLIMB:34-114s | CRUISE:114-266s |
DESCENT:266-334s | LANDING:334-380s
[PASS] ON_GROUND altitude = 0
[PASS] CRUISE altitude ~9000m
[PASS] TAKEOFF altitude > 0
[PASS] CLIMB altitude > 500m
[PASS] DESCENT altitude < 9000m
[PASS] LANDING altitude < 500m

+-----+
? Flight Phase ? Target Speed ?
+-----+

[PASS] ON_GROUND speed = 0
[PASS] TAKEOFF speed = 75 m/s
[PASS] CRUISE speed = 232 m/s (451 kts)
[PASS] CLIMB speed > 75 m/s
[PASS] CLIMB speed <= 220 m/s

+-----+
? Flight Phase ? Target Pitch & Roll ?
+-----+

[PASS] ON_GROUND pitch=0
[PASS] TAKEOFF pitch=12
[PASS] CLIMB pitch=8
[PASS] CRUISE pitch=2
[PASS] DESCENT pitch=-3
[PASS] LANDING pitch=5
[PASS] TAKEOFF roll=0
[PASS] LANDING roll=0
[PASS] CRUISE roll=2

+-----+
? Crash Scenario ? Names ?
+-----+

[PASS] CRASH_NONE name
[PASS] CRASH_STALL name
[PASS] CRASH_OVERSPEED name
[PASS] CRASH_NOSE_DIVE name
[PASS] CRASH_SPIRAL name
[PASS] CRASH_ENGINE_FAILURE name
[PASS] CRASH_TERRAIN name

+-----+
? Crash Scenario ? CrashInfo ?
+-----+

[PASS] CrashInfo.type = STALL
[PASS] CrashInfo.trigger_time = 200
[PASS] CrashInfo.impact_time = trigger+30
[PASS] CrashInfo.description not empty
[PASS] CrashInfo.cause not empty
[PASS] CrashInfo NONE type

+-----+
? Crash Scenario ? Altitude Effects ?
+-----+

[PASS] STALL reduces altitude
[PASS] OVERSPEED reduces altitude
[PASS] NOSE_DIVE reduces altitude
[PASS] SPIRAL reduces altitude
[PASS] ENGINE_FAILURE reduces altitude
[PASS] TERRAIN reduces altitude
[PASS] STALL altitude >= 0
[PASS] OVERSPEED altitude >= 0
[PASS] TERRAIN faster than ENGINE_FAILURE
[PASS] SPIRAL faster than STALL
[PASS] At trigger moment, altitude unchanged


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+-----+
? Circular Buffer ? Chronological Order?
+-----+
[BLACKBOX] Initialized (capacity: 200)
[PASS] 50 records returned
[PASS] Records are in chronological order

+-----+
? Circular Buffer ? Re-initialisation ?
+-----+
[BLACKBOX] Initialized (capacity: 200)
[PASS] 20 records before reinit
[BLACKBOX] Initialized (capacity: 200)
[PASS] 0 records after reinit
[PASS] get_ordered_records returns NULL when empty

+-----+
? Altitude Sensor ? Normal Flight ?
+-----+
[PHASES] Duration=380s | ON_GROUND:0-15s | TAKEOFF:15-34s | CLIMB:34-114s | CRUISE:114-266s |
DESCENT:266-334s | LANDING:334-380s
[PASS] ON_GROUND altitude = 0
[PASS] CRUISE altitude near 9000m
[PASS] Altitude clamped to [0,10000]

+-----+
? Altitude Sensor ? Crash Effects ?
+-----+
[PASS] Altitude decreases during crash
[PASS] Altitude never negative during crash

+-----+
? Speed Sensor ? Normal Flight ?
+-----+
[PASS] ON_GROUND speed = 0
[PASS] After 30 ticks CRUISE speed ~232 m/s (451 kts)
[PASS] Speed decreases when crashed
[PASS] Speed never negative when crashed

+-----+
? Speed Sensor ? Unit Conversion ?
+-----+
[PASS] 0 m/s = 0 kts
[PASS] 1 m/s = 1.944 kts
[PASS] 220 m/s = ~427 kts
[PASS] 100 m/s = ~194 kts

+-----+
? Pitch Sensor ? Normal Flight ?
+-----+
[PASS] Pitch within sensor limits [-10,15]
[PASS] NOSE_DIVE crash pitch = -45 deg

+-----+
? Roll Sensor ? Normal Flight ?
... (output truncated to 300 lines)

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